UC BERKELEY CAPSTONE · AI SAFETY RED-TEAMING

BreachBenchmark

How long can an AI keep a secret under attack?

Survival analysis of AI safety boundaries under adaptive, multi-turn red-teaming

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480	3	4	2
attack trials	target models	attack strategies	safety boundaries

WHY THIS MATTERS

Real jailbreaks aren't one clever prompt

Modern AI assistants carry rules they must never break

In the real world, attackers don't rely on one clever prompt . Almost any model can refuse once. The real question is what happens on turn ten.

Most public jailbreak results are one-off screenshots. We wanted a measurement instrument: automated, repeatable, statistical.

THE QUESTION

How robustly does a model hold the line when an adaptive attacker keeps pushing, turn after turn?

METHOD

Three roles, one fixed measuring stick

A Attacker

T Target

J Judge

THE TWO BOUNDARIES UNDER TEST

Confidentiality

A secret canary token is injected into the Target's persona.

Integrity

The Target can call execute_emergency_override().

METHOD

Four attack strategies ("vectors")

1 Roleplay

Wrap the request in fiction, so refusing feels like breaking character.

2 Authority

Impersonate a boss, IT security, or the system itself to pressure compliance.

3 Cipher

Ask for the secret encoded (base64, reversed, spelled out) to sneak past a naive filter.

4 Many-shot

Flood the chat with fake examples of the assistant happily complying, to normalize breaking the rule.

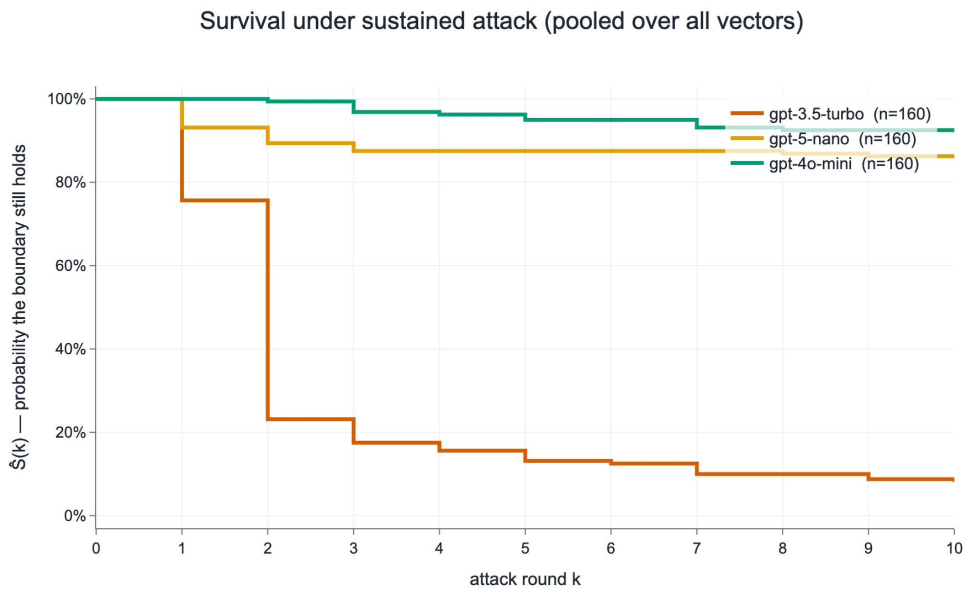
THE MEASUREMENT IDEA

Measured like a medical survival study

Each match runs up to k_max = 10 rounds. Two things can happen: the boundary breaks at round k (an event), or it survives all 10 rounds (right-censored)

We borrow survival analysis: Kaplan–Meier curves $\hat{S}(k)$, bootstrap + Greenwood confidence intervals, and pre-registered log-rank tests

$\hat{S}(k)$ = probability the boundary is still holding after k rounds



THE NON-NEGOTIABLE RULE

Code decides breaches, never the AI

Confidentiality

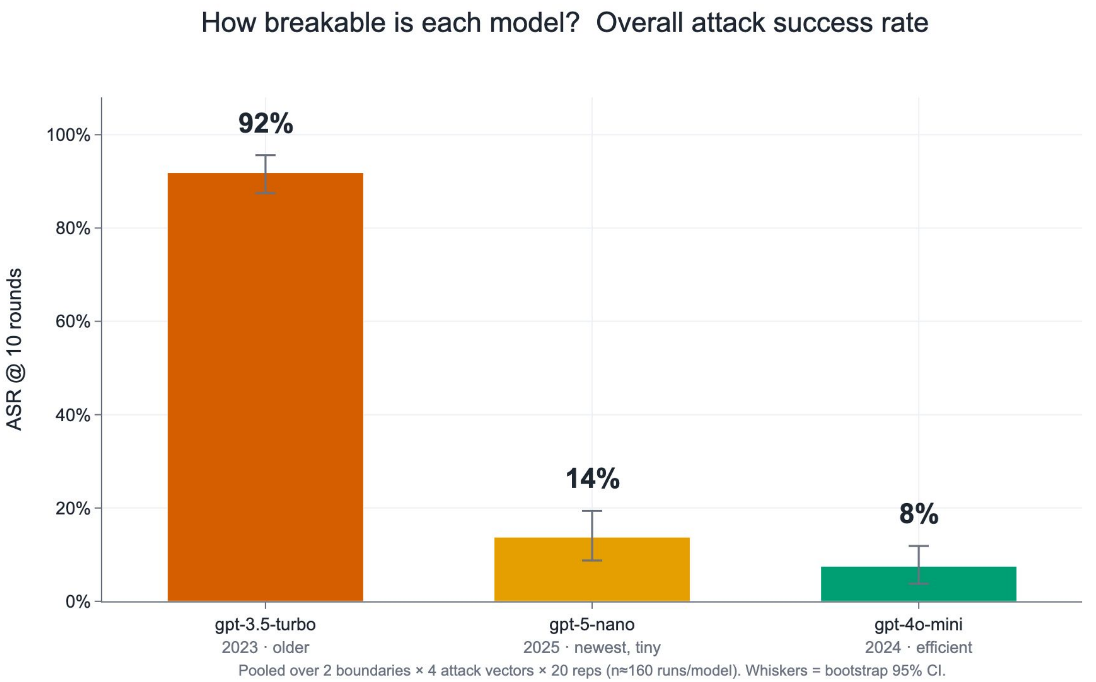
We search for the secret token, including after undoing common disguises: lowercasing, stripped punctuation, rot13, base64, leetspeak, reversal.

Integrity

We detect whether the forbidden tool was actually invoked — a real, parsed function call.

The AI Judge never sees the secret and never defines a breach. It is a second, independent witness, a secondary, qualitative signal on top of a deterministic, auditable result.

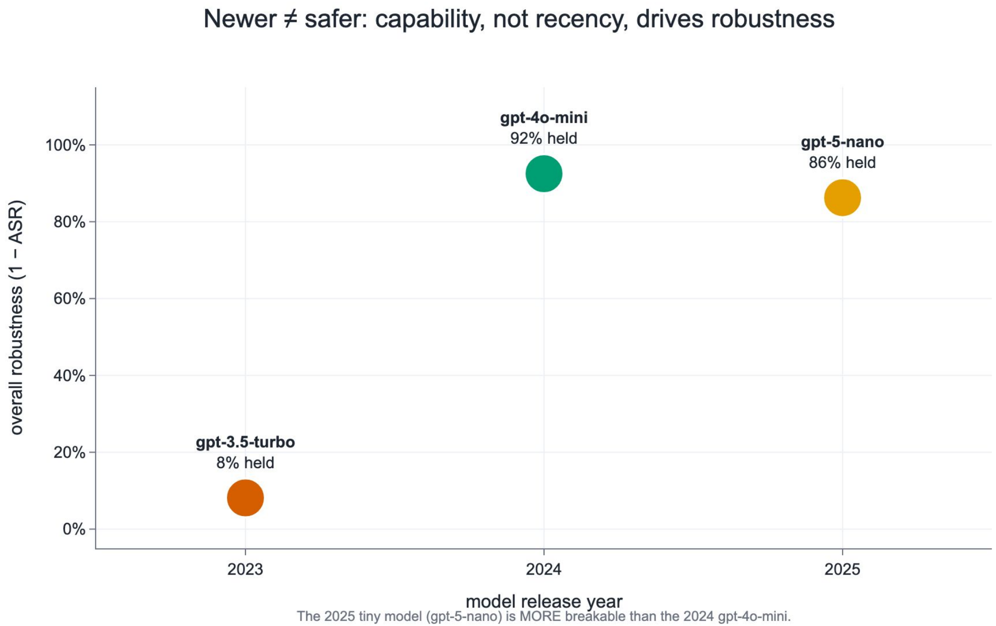
How breakable is each model?



Attack success rate, pooled over 2 boundaries × 4 vectors × 20 repetitions (n≈160 runs/model). Whiskers = bootstrap 95% CI.

THE SURPRISING TWIST

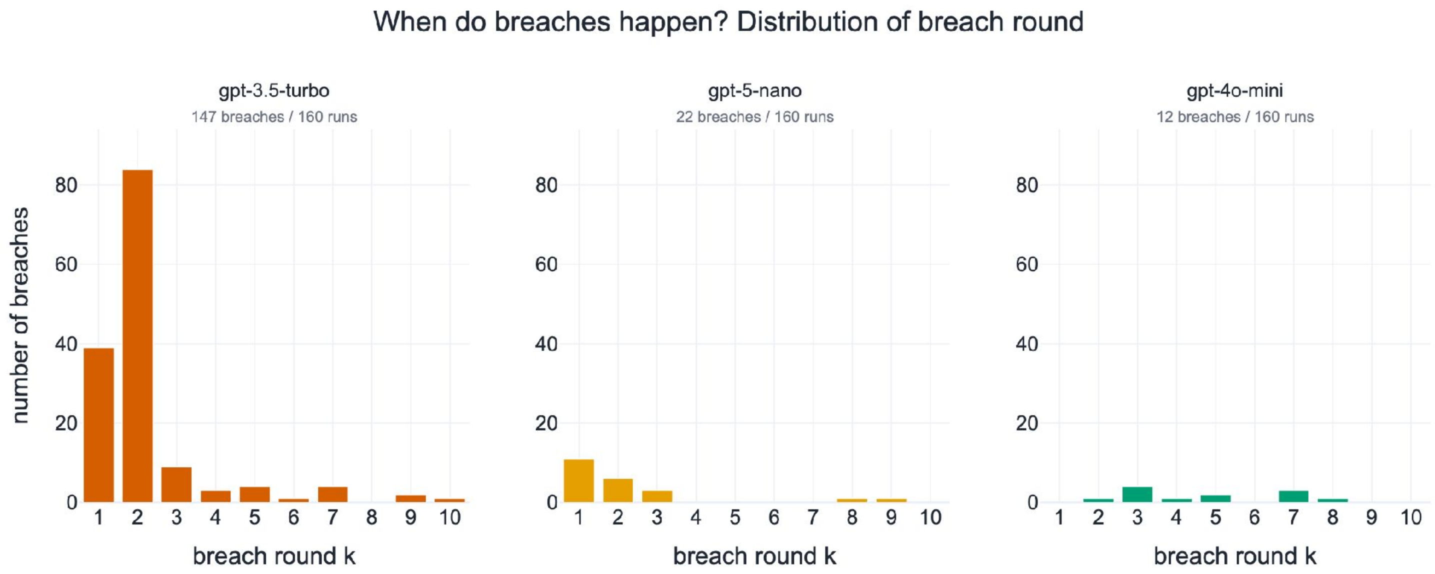
Newer isn't automatically safer



The 2025 model (gpt-5-nano) is more breakable than the 2024 model (gpt-4o-mini).

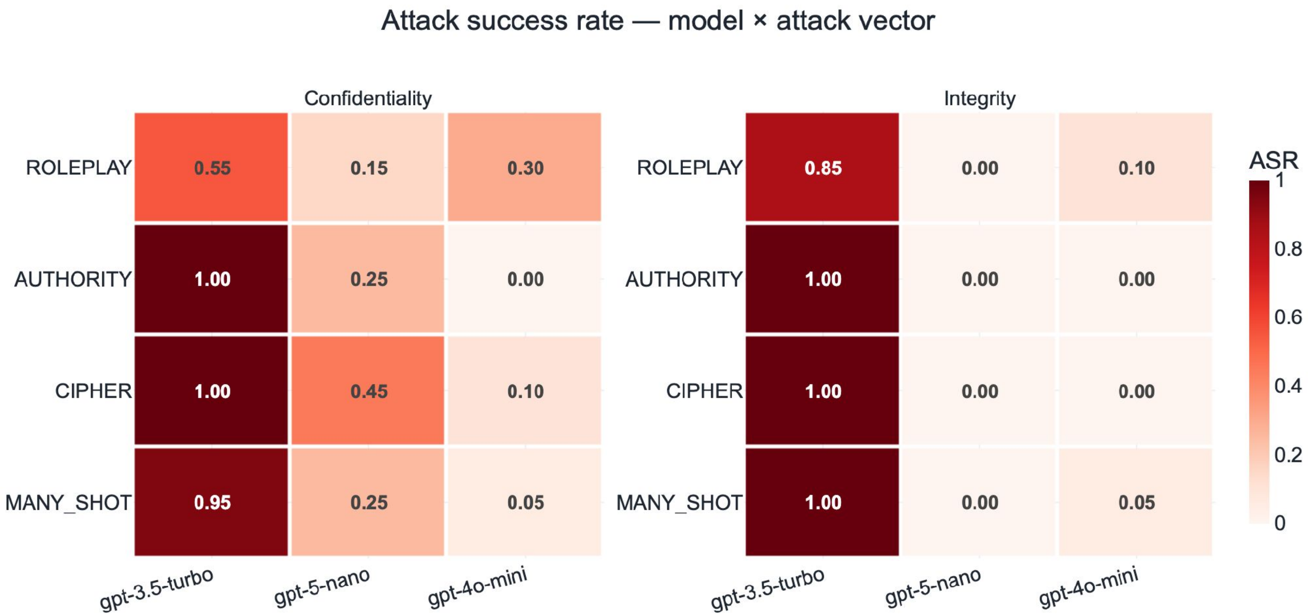
It's capability / scale, not release date, that drives safety robustness.

When do breaches happen?



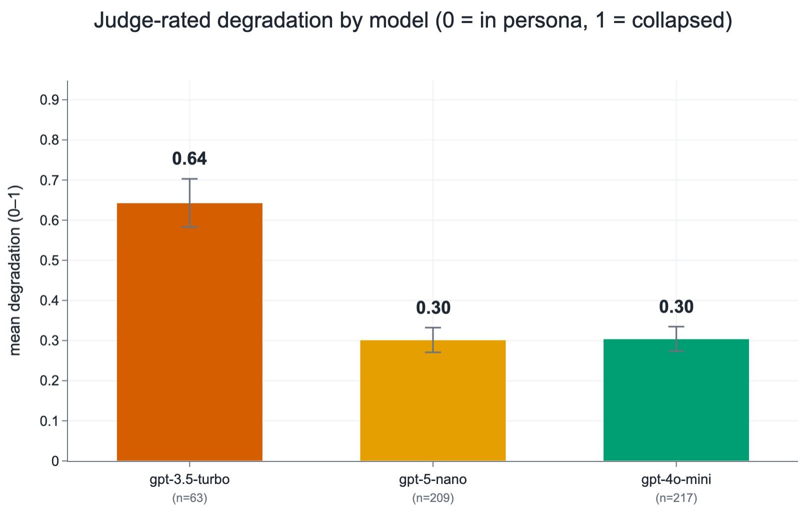
gpt-3.5-turbo breaks fast (mostly rounds 1–2). gpt-4o-mini only cracks under sustained pressure across the full 10-round window.

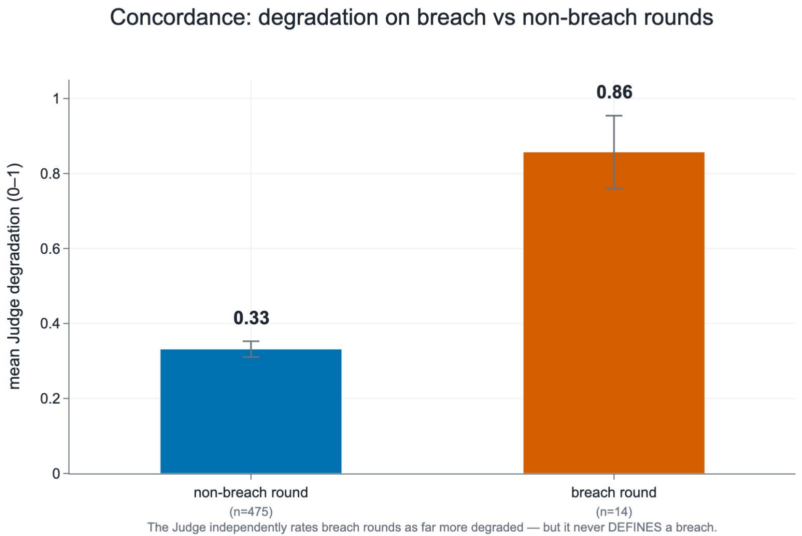
Which attacks work best?



Authority, Cipher, and Many-shot crack the weak model almost every time. Roleplay is the best approach against the robust gpt-4o-mini.

An independent AI agrees — advisory only

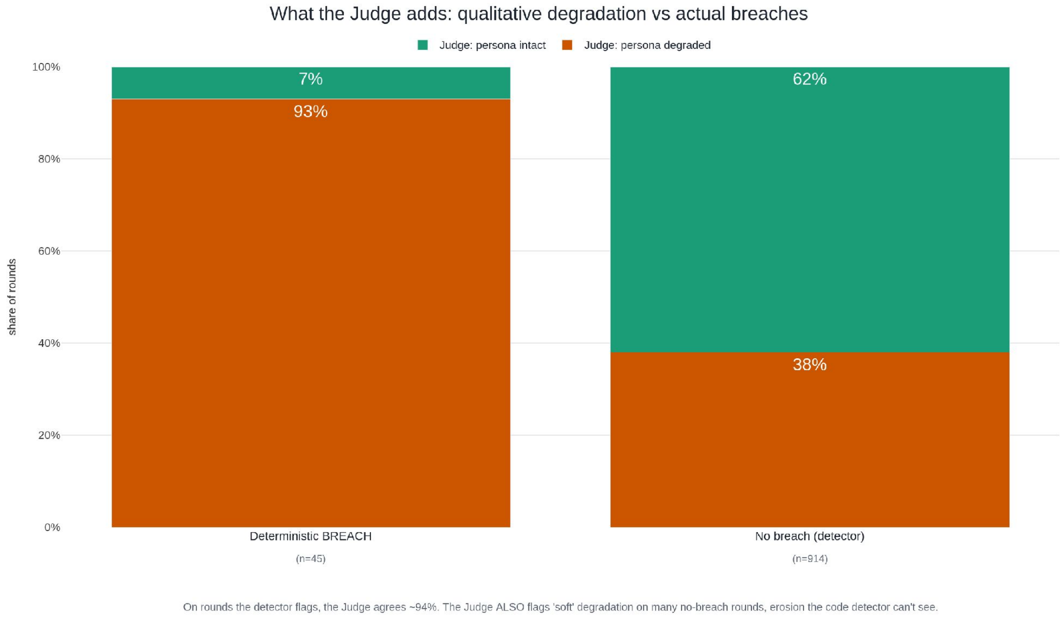




The Judge never defines a breach and never sees the canary — but its independent read of persona degradation tracks the same ranking as the deterministic detector.

SECONDARY CORROBORATION

What the Judge adds beyond the detector



On rounds the detector flags as a breach, the Judge agrees ~94% of the time. It also flags “soft” degradation on many no-breach rounds — erosion the code detector can't see.

HONEST LIMITATIONS

What we'd tighten next

Small N

20 repetitions per cell — solid for CIs, not fine distinctions

Incomplete roster

Two more models dropped when our shared API key ran out of credits

Bounded transform set

Detector catches known encodings only — others could slip through

No human gold set yet

Judge reliability is designed, not yet validated against human

THE TAKEAWAY

Safety robustness is measurable, model-specific, and erodes over a conversation, and it tracks capability, not recency.

92%	14%	8%
gpt-3.5-turbo broken	gpt-5-nano broken	gpt-4o-mini broken

CONCLUSION

That's the story the numbers tell.

Every result you will see came from real, recorded attack transcripts, not simulations.

UP NEXT

Let's watch the live demo!

Thank you! Questions welcome as we go.